Discrete Latent Sequence Generation for 12-Lead ECG Reconstruction

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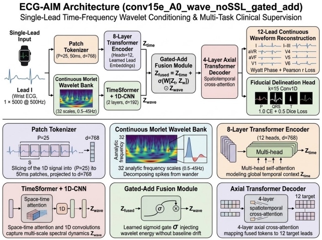
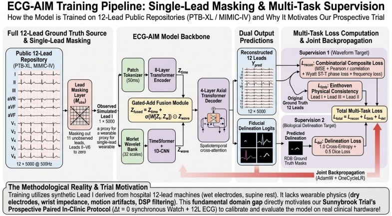
Clinical Background & Motivation

- **12-Lead Gold Standard**: Standard clinical cardiology relies on 10 electrodes capturing 3D cardiac electrical dipoles across frontal and horizontal planes, critical for diagnosing myocardial infarction, ischemia, and chamber hypertrophy.
- **Ambulatory Wearable Reality**: Consumer smartwatches (Lead I) and patch monitors (Lead II) provide continuous at-home monitoring but are blind to 10 of 12 spatial vectors

- **Novel Generative Framework**: Formulate **ECG-AIM**, a discrete latent sequence generation framework combining multi-resolution Morlet wavelets with spatiotemporal axial attention to reconstruct 12 leads from limited inputs.
- **Factorial 2^4 Loss Benchmarking**: Systematically dissect the contributions of amplitude, correlation, distribution, and slope constraints across 3 model paradigms (U-Net, MultiScale VAE

Methods: Architecture, Multi-Objective Loss & Training Pipeline





1. Dual-Branch Model Architecture

Patch Tokenizer: 50 ms patches mapped to $d = 768$ temporal tokens via an 8-layer Transformer encoder.
Morlet Wavelet Bank: 32 analytic scales (0.5–45 Hz) extracts multi-resolution time-frequency scalograms.
Gated-Add Fusion: $Z_{\text{fused}} = Z_{\text{time}} + \sigma(W[Z_t, Z_w]) \odot \text{Proj}(Z_w)$.
Axial Decoder: Spatiotemporal cross-attention predicting 12-lead waveforms and explicit P-QRS-T masks.

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + 0.5\mathcal{L}_{\text{corr}} + 0.1\mathcal{L}_{\text{MMD}} + 0.05\mathcal{L}_{\Delta} + \lambda\mathcal{L}_{\text{limb}}$$

Kirchhoff Limb Prior: $\mathcal{L}_{\text{limb}} = \|y_{\text{II}} - (y_{\text{I}} + y_{\text{III}})\|_2$

Results and Conclusion: Clinical Precision & Hardware Translation

Endpoint Dimension	Baseline U-Net	Stanford 3DRECON	ECG-AIM (Ours)	Clinical Impact
Precordial Chest Mean r	0.570	0.732	**0.776**	$\Delta = +0.206$ ($p < 10^{-300}$)
QRS Duration MAE	13.85 ms	10.26 ms	**8.0 ms (Median)**	Sub-10ms clinical gate
Conduction Delay (>120ms)	90.7%	94.3%	**94.3%**	aOR $= 1.84$ ($p < 10^{-17}$)
Sokolow-Lyon LVH Voltage	1.13 mV	0.63 mV	**0.72 mV**	$r = 0.714$ chamber concordance
EchoNext Structural Disease	68.8%	**46.5% (Collapse)**	**71.6%**	Preserves echo pathology

Leads I, II, and V_2 lock the 3 orthogonal dipoles (X, Y, Z), achieving precordial Pearson r = **0.980–1.000** and preserving anterior ST elevation.

Prior state-of-the-art (Stanford 3DRECON) collapsed to **46.5%** on downstream EchoNext structural heart disease. ECG-AIM maintains **71.6%** by preventing amplitude regression.